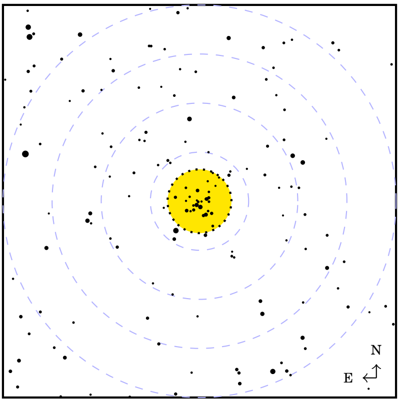
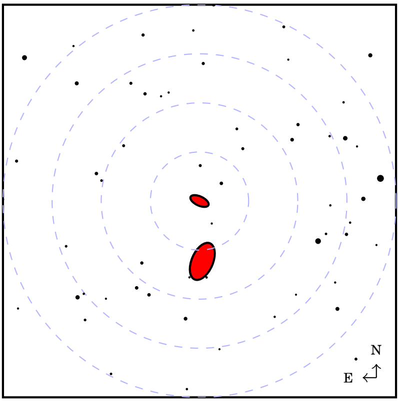
Pocket Finder-Chart Atlas





Correct-Image Version

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Pocket Finder-Chart Atlas

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Correct-image version of 2022-03-26

Contents

Preface

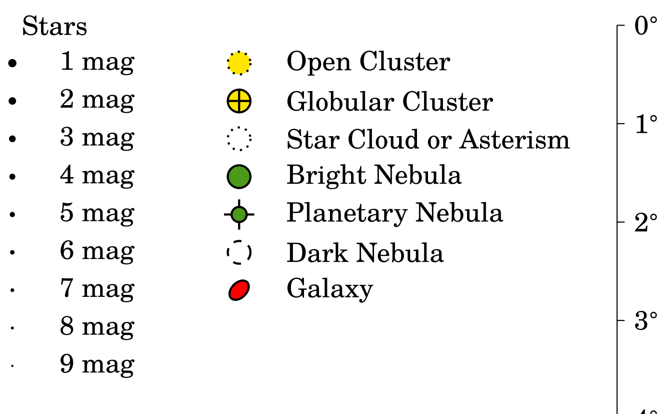
This is an atlas of finder charts for the Messier, Caldwell, and Astronomical League Urban Observing Program deep-sky objects.

The charts have a field of 4 degrees at a scale of 13 mm per degree. Circles show fields with diameters of 1, 2, 3, and 4 degrees. The charts show stars to magnitude 9.5, and represent deep-sky objects following the current convention. The correspondence between the magnitude of a star and its size on the charts is designed to better match the view in a telescope eyepiece.

There are versions of the atlas with inverted-image and correct-image charts. This version has correct-image charts, with north up and east to the left, corresponding to the view through a refractor or Cassegrain telescope either without a diagonal or with a correct-image diagonal.

The size and binding of this atlas are designed for convenience at the telescope.

Chart Legend



I created this atlas to assist me as I observed under the light-polluted skies of Mexico City with a 70 mm f/6 refractor. I don't use a conventional finder with this telescope, but instead star-hop using a 32 mm eyepiece with a true field of about 4 degrees. Under my usual observing conditions, very few objects are immediately obvious, and the charts tell me when I've successfully reached the desired field and where to focus my efforts.

The obvious question is, but why not just use use an all-sky atlas? I do indeed use the *Pocket Sky Atlas* for star-hopping, but I find that for confirming a field none of my all-sky atlases combine adequate depth with convenience at the telescope, and none really give me the sense of what I see through the eyepiece.

To some degree, the relation between these charts and the *Pocket Sky Atlas* is similar to the relation between the larger-scale and smaller-scale charts in *The Observer's Sky Atlas*. The smaller-scale charts are for star-hopping to the field and the larger-scale charts are to confirm the field and locate the object. I don't use *The Observer's Sky Atlas* at the telescope, because the smaller-scale charts are too shallow and too narrow for my needs, the large-scale charts aren't inverted, and the binding is inconvenient.

The positions and magnitudes of stars are taken from the Tycho-2 catalog, supplemented by the Hipparcos catalog for the very brightest stars. The positions, types, and sizes of deep sky objects are taken from the CDS SIMBAD database, supplemented by information from the NASA/IPAC Extragalactic Database (NED), O'Meara's *Deep-Sky Companions* volumes, and the Astronomical League's Urban Observing Program list.

One flaw of these charts is that all deep-sky objects are drawn as either ellipses (galaxies and star clouds) or circles (everything else). This means, for example, that they do not correctly show the shapes of irregular nebulae such as M42 and M43.

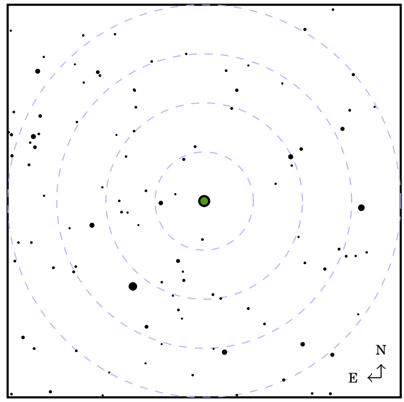
Test Objects

M1
NGC 1952
Crab Nebula

RA : 5.6^h	PSA : 14	Type: BN
Dec: $+22°$	TOSA: E3	Mag : 8.0
Con: Tau	TCSA: 3/9	Size : $6' \times 4'$

ζ Tau (3.0) $\rightarrow$ 1.1° NW $\rightarrow$ M1

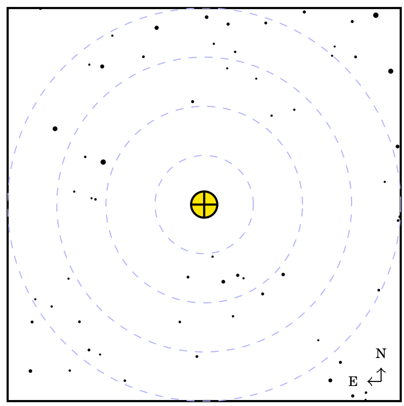
See: TMO (p. 43)



M2

RA : ^{h}	PSA :	Type:
Dec: °	TOSA:	Mag :
Con:	TCSA:	Size :

See:



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